	Tutorial -2 [Calculus]
1	Find the points of intersection of the curves given by the polar coordinate equations $r = \sin\theta, \quad r = 1 + \sin\theta$ $r = 1 + \cos\theta, \quad r = 1 - \cos\theta$ $r = 1 + \sin\theta, \quad r = 1 - \sin\theta$ $r = 1 + \sin\theta, \quad r = -1 + \sin\theta$ $r = 1 + \cos\theta, \quad r = -1 + \cos\theta$
2	Find polar equations for the circles . Sketch each circle in the coordinate plane and label it with both its Cartesian and polar equations. $x^2 + y^2 + 5y = 0$ $\quad$ $x^2 + y^2 - 2y = 0$ $x^2 + y^2 - 3x = 0$ $\quad$ $x^2 + y^2 + 4x = 0$
3	Find the areas of the regions in the polar coordinate plane described in Enclosed by the limaçon $r = 2 - \cos\theta$ Enclosed by one leaf of the three-leaved rose $r = \sin 3\theta$ Inside the "figure eight" $r = 1 + \cos 2\theta$ and outside the circle $r = 1$
4	Find the lengths of the curves given by the polar coordinate equations $r = -1 + \cos\theta$ $r = 2\sin\theta + 2\cos\theta, \quad 0 \le \theta \le \pi/2$ $r = 8\sin^3(\theta/3), \quad 0 \le \theta \le \pi/4$ $r = \sqrt{1 + \cos 2\theta}, \quad -\pi/2 \le \theta \le \pi/2$

5	Find the volume of the solid generated by revolving the region enclosed by the ellipse $9x^2 + 4y^2 = 36$ about **(a)** the x-axis, **(b)** the y-axis. The "triangular" region in the first quadrant bounded by the x-axis, the line $x = 4$, and the hyperbola $9x^2 - 4y^2 = 36$ is revolved about the x-axis to generate a solid. Find the volume of the solid.
6	Show that the equations $x = r\cos\theta$, $y = r\sin\theta$ transform the polar equation $$r = \frac{k}{1 + e\cos\theta}$$ into the Cartesian equation $$(1 - e^2)x^2 + y^2 + 2kex - k^2 = 0.$$
7	Find the points on the parabola $x = 2t$, $y = t^2$, $-\infty < t < \infty$, closest to the point $(0, 3)$.
8	Find points on the cardioid $r = a(1 + \cos\theta)$ where the tangent line is **(a)** horizontal, **(b)** vertical. Show that parabolas $r = a/(1 + \cos\theta)$ and $r = b/(1 - \cos\theta)$ are orthogonal at each point of intersection $(ab \neq 0)$. Find the angle at which the cardioid $r = a(1 - \cos\theta)$ crosses the ray $\theta = \pi/2$.
9	Find the angle at which the parabolas $r = 1/(1 - \cos\theta)$ and $r = 1/(1 - \sin\theta)$ intersect in the first quadrant. The equation $r^2 = 2\csc 2\theta$ represents a curve in polar coordinates. **a.** Sketch the curve. **b.** Find an equivalent Cartesian equation for the curve. **c.** Find the angle at which the curve intersects the ray $\theta = \pi/4$.

10	Find the points of intersection of the parabolas $$r = \frac{1}{1 - \cos\theta} \quad \text{and} \quad r = \frac{3}{1 + \cos\theta}$$ and the angles between their tangents at these points.
11	Find the point on the curve $y = e^x$ where the curvature is greatest.
12	**Radius of curvature** Show that the radius of curvature of a twice-differentiable plane curve $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$ is given by the formula $$\rho = \frac{\dot{x}^2 + \dot{y}^2}{\sqrt{\ddot{x}^2 + \ddot{y}^2 - \ddot{s}^2}}, \quad \text{where} \quad \ddot{s} = \frac{d}{dt}\sqrt{\dot{x}^2 + \dot{y}^2}.$$
13	Find $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$, κ, and τ as functions of t if $\mathbf{r}(t) = (\sin t)\mathbf{i} + \left(\sqrt{2}\cos t\right)\mathbf{j} + (\sin t)\mathbf{k}$.
14	The position of a particle moving in space at time $t \geq 0$ is $$\mathbf{r}(t) = 2\mathbf{i} + \left(4\sin\frac{t}{2}\right)\mathbf{j} + \left(3 - \frac{t}{\pi}\right)\mathbf{k}.$$ Find the first time $\mathbf{r}$ is orthogonal to the vector $\mathbf{i} - \mathbf{j}$.
15	Find the lengths of the curves in $$\mathbf{r}(t) = (2\cos t)\mathbf{i} + (2\sin t)\mathbf{j} + t^2\mathbf{k}, \quad 0 \leq t \leq \pi/4$$ $$\mathbf{r}(t) = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + 2t^{3/2}\mathbf{k}, \quad 0 \leq t \leq 3$$
16	In $\mathbf{r}(t)$ find $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$, κ, and τ at the given value of t. $$\mathbf{r}(t) = \frac{4}{9}(1 + t)^{3/2}\mathbf{i} + \frac{4}{9}(1 - t)^{3/2}\mathbf{j} + \frac{1}{3}t\mathbf{k}, \quad t = 0$$ $$\mathbf{r}(t) = (e^t \sin 2t)\mathbf{i} + (e^t \cos 2t)\mathbf{j} + 2e^t\mathbf{k}, \quad t = 0$$

17	In Exercises find the partial derivative of the function with respect to each variable. $g(r, \theta) = r\cos\theta + r\sin\theta$ $f(x, y) = \frac{1}{2}\ln(x^2 + y^2) + \tan^{-1}\frac{y}{x}$ $f(R_1, R_2, R_3) = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ $h(x, y, z) = \sin(2\pi x + y - 3z)$
18	Find the second-order partial derivatives of the functions $g(x, y) = y + \frac{x}{y}$ $f(x, y) = x + xy - 5x^3 + \ln(x^2 + 1)$ $f(x, y) = y^2 - 3xy + \cos y + 7e^y$ $g(x, y) = e^x + y\sin x$
19	Find dw/dt at $t = 0$ if $w = \sin(xy + \pi), x = e^t$, and $y = \ln(t + 1)$. Find dw/dt at $t = 1$ if $w = xe^y + y\sin z - \cos z, x = 2\sqrt{t}$, $y = t - 1 + \ln t$, and $z = \pi t$. Find $\partial w/\partial r$ and $\partial w/\partial s$ when $r = \pi$ and $s = 0$ if $w = \sin(2x - y)$, $x = r + \sin s, y = rs$. Find $\partial w/\partial u$ and $\partial w/\partial v$ when $u = v = 0$ if $w = \ln\sqrt{1 + x^2} - \tan^{-1} x$ and $x = 2e^u \cos v$.

20	**Derivative in velocity direction** Find the derivative of $f(x, y, z) = xyz$ in the direction of the velocity vector of the helix $$\mathbf{r}(t) = (\cos 3t)\mathbf{i} + (\sin 3t)\mathbf{j} + 3t\mathbf{k}$$ at $t = \pi/3$. **Maximum directional derivative** What is the largest value that the directional derivative of $f(x, y, z) = xyz$ can have at the point $(1, 1, 1)$?
21	In Exercises find parametric equations for the line that is tangent to the curve of intersection of the surfaces at the given point. Surfaces: $x^2 + 2y + 2z = 4, \quad y = 1$ Point: $(1, 1, 1/2)$ Surfaces: $x + y^2 + z = 2, \quad y = 1$ Point: $(1/2, 1, 1/2)$
22	Test the functions in Exercises 6 for local maxima and minima and saddle points. Find each function's value at these points. $f(x, y) = x^2 - xy + y^2 + 2x + 2y - 4$ $f(x, y) = 5x^2 + 4xy - 2y^2 + 4x - 4y$ $f(x, y) = 2x^3 + 3xy + 2y^3$ $f(x, y) = x^3 + y^3 - 3xy + 15$ $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2$

23	**Extrema on a circle** Find the extreme values of $f(x, y) = x^3 + y^2$ on the circle $x^2 + y^2 = 1$. **Extrema on a circle** Find the extreme values of $f(x, y) = xy$ on the circle $x^2 + y^2 = 1$. **Extrema in a disk** Find the extreme values of $f(x, y) = x^2 + 3y^2 + 2y$ on the unit disk $x^2 + y^2 \leq 1$. **Extrema in a disk** Find the extreme values of $f(x, y) = x^2 + y^2 - 3x - xy$ on the disk $x^2 + y^2 \leq 9$. **Extrema on a sphere** Find the extreme values of $f(x, y, z) = x - y + z$ on the unit sphere $x^2 + y^2 + z^2 = 1$.
24	Let $w = f(r, \theta), r = \sqrt{x^2 + y^2}$, and $\theta = \tan^{-1}(y/x)$. Find $\partial w/\partial x$ and $\partial w/\partial y$ and express your answers in terms of r and θ. Let $z = f(u, v), u = ax + by$, and $v = ax - by$. Express z_x and z_y in terms of f_u, f_v, and the constants a and b. If a and b are constants, $w = u^3 + \tanh u + \cos u$, and $u = ax + by$, show that $$a\frac{\partial w}{\partial y} = b\frac{\partial w}{\partial x}.$$
25	**Extrema on curve of intersecting surfaces** Find the extreme values of $f(x, y, z) = x(y + z)$ on the curve of intersection of the right circular cylinder $x^2 + y^2 = 1$ and the hyperbolic cylinder $xz = 1$. **Minimum distance to origin on curve of intersecting plane and cone** Find the point closest to the origin on the curve of intersection of the plane $x + y + z = 1$ and the cone $z^2 = 2x^2 + 2y^2$.